

Touch the Collections: A 3D UI Prototype in DSpace-7

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ABSTRACT

Institutions have focused on the digitization process, metadata workflow, and digital infrastructure development for digital librarianship in recent decades. The undergoing digital shift that is accelerated by the pandemic crisis has signified the importance of visualizing digital collections because digital presence is no longer a preview of the physical collection; instead, the digital presence has become the primary access point. Replacing the heavily text-based browsing features, visualization and interactive design in digital repositories are used to facilitate communication, presentation, discovery, and comprehension of digital and digitized heritage. This project explores and demos the first 3D user interface that visually outlines and interactively exhibits communities and collections in DSpace-7 and discusses the applicable scenarios. The integration of the developed modules will be released as open sources on GitHub once completed.

CCS CONCEPTS

• Human-centered computing ~ Visualization ~ Visualization systems and tools ~ Visualization toolkits • Information systems ~ Information systems applications ~ Digital libraries and archives • Applied computing ~ Education ~ Digital libraries and archives

KEYWORDS

Open Repository, DSpace-7, Data Visualization, 3D Prototype

1 INTRODUCTION

Cultural heritage institutions have been investing in digital systems, workflow efficiency, content quality, and discoverability to address the increasing demand for accessing digital and digitized assets since the 1990s [6]. Only a few efforts have been devoted to data visualization of digital collections until the most recent decade. As a result, several visualization prototypes and cases have been designed and implemented to enhance user experiences in digital repository systems [5]. Although a substantial increase in the use of digital and digitized materials has been witnessed, the digital shift in the library world has only accelerated once the global pandemic confronted higher education in 2020, when print collections became inaccessible and unreachable resources for those who are impacted and quarantined. Since then, digital content has suddenly shifted from

being an alternative to physical heritage assets to an immediate access point [3].

Users often land on digital objects through generic Internet search engines or library-implemented discovery systems without any contextual knowledge of what repositories, communities, or collections the landed objects are included in [5]. Nevertheless, digital collection interfaces are fulfilled by text-based browsing features and bullet-based relational frames; users are restricted by user-unfriendly elements from further understanding and discovery. Similar to the humanists' examining how technologies are being implemented and impacting liberal arts, digital library practitioners can explore how digital tools could be used then shaping users' understanding of digital librarianship [2]; and starting there, data visualization techniques should be treated and eventually to become a common feature of most search interfaces.

Indeed, visualization has long dominated the human perception of information and knowledge, which had been used to convey essential information [4], and has become an integral part of scholarly communication and knowledge economies [1]. The attractiveness of data visualization stems from its ability to simplify complex data, enhance discovery, support exploration, improve content analysis, and most importantly, engage the audience in the communication process to facilitate comprehension [6]; wherefore we, the authors of this paper, are attracted to participate and contribute to the visualization culture, exploring to integrate such a visualization technology to the newest DSpace-7 open repository system.

2 INTEGRATING 3D TO DSPACE-7

Three-dimensional data visualization technology has been established in many industries; the technology includes three stages: scene, geometry, and rendering. It can present qualitative and quantitative information about objects and the relationships among objects. 3D visualization is effective in presentations and communications with digital assets, adding new perspectives in discovering and exploring digital collections, telling better stories by rendering data in a more comprehensive format, and highlighting developmental trends. 3D visualization technologies can improve or analyze architectural relationships to design and imitate the data model and structures. Users can perform the actions like touching, zooming, dragging, panning, orbiting,

flying, rotating, and translating, which make 3D visualization powerful in presentations and communications.

2.1 Development & Integration

In our experiment, we used Threejs, an open-source JavaScript library, to display the graphics, 3D, and 2D objects on the web browser, as well as its add-ons and other libraries. To make it work with DSpace Angular, we installed three different libraries and imported them to the custom components. The 3D environment is implemented on a canvas. The geometry shapes in Threejs has plane, cube, sphere, torus, and more. We chose spheres to simulate the DSpace data model: communities, collections, and items. For those related objects, e.g., an article and its parent collection, are connected by a line. The virtual environment mimicked a universe with the galaxy system (See Figure 1). Other animation effects were also added to provide a good user experience.



Figure 1: Prototype of 3D UI on DSpace-7

2.2 Descriptive Operation

In doing this, we are trying to navigate the DSpace-7 development community to the “touch” community, enabling the interactive feature for users to explore, discover, and comprehend digital collections at the level of communities, collections, items, and profiles in a virtual environment.

As illustrated in Figure 1, all 3D objects are mapped according to the data. The initial 3D scene contains a sphere in the center representing the repository site, surrounded by other spheres representing the top communities in the repository. Clicking or touching one of the community spheres will be zoomed to the center of the community, illustrated by two information panels. One panel contains information and statistics on the selected community, and the other panel shows relationships between sub-communities and sub-collections. Sub-communities and sub-collections are also presented as 3D satellites surrounding the top community. Selecting a collection or items within the collection will generate connection lines between each item and the collection. Users can browse, search, and filter content in the 3D

scene. In short, our goal is to develop a 3D UI in which users can perform many activities as they do in regular DSpace repositories.

2.3 Application Scenarios

There are several places in which we think this new development can be implemented in DSpace-7, including:

1. Community page: Use 3D objects to visualize the structure of communities, sub-communities, and collections.
2. Collection page: Illustrate a collection surrounded by items with lines connected to each item.
3. Item page: Illustrate a single digital object surrounded by authors and other entities.
4. Search result and filter page: Present searched items in the 3D scene of a result page.

In addition, there are several scenarios we believe the development can contribute to, including:

1. Presentation: Visualize the information when debriefing to the administration or presenting for financial support.
2. Scholarly communication: Demo the collections and digital initiatives efforts to stakeholders on/off campus.
3. Upskilling: Accumulate knowledge and skill level for 3D development in digital repositories.

3 CONCLUSION

Data visualization is a blended field of science and art. It conveys information and concepts differently to help users view data in an intuitive and straightforward way so that users can grasp concepts, relationships, and patterns and drill down into data for more details. Adding a different perspective to view the digital collections in the virtual world engages users in a new repository landscape. The 3D prototype application unlocks new possibilities to design innovation for information repositories, and we hope this work will enrich discovery and exploration.

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